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Data-driven decomposition of seasonal time series

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Abstract

This paper discusses decomposition of seasonal time series based on locally weighted regression. With polynomials and trigonometric functions as local regressors a time series is decomposed into unobserved components. The asymptotic behaviour of the proposed estimators are investigated. The focus is on the development of a data-driven procedure. For the selection of bandwidths a mean averaged squared error criterion is used with a bootstrap variance estimator. The selection of the optimal polynomial order is also considered. For the smooth trend-cyclical component also the first derivative, which gives information about instantaneous changes of the trend-cyclical when applied to macro-economic data, is estimated. Application to a German macro-economic time series is presented as illustration. Differences between the proposed method and the other well-known ones based on parametric models are briefly discussed.
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1. Introduction

Local regression has a long tradition in smoothing and seasonal decomposition of time series (see e.g. Macaulay, 1931). For a historical review of unobserved component models and seasonal adjustment in time series see Eurostat (1998) or Heiler (1995a, b). Among economic agencies the results of such procedures are broadly used for the assessment and judgement of business cycle behaviour from economic data. Heiler (1966, 1970) developed a decomposition procedure based on local regression with polynomials and trigonometric functions as local regressors. This idea became the basis of the so-called Berlin Method (Berliner Verfahren, BV), which in its fourth

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version (BV 4) is being applied by the German Federal Statistical Office since 1983. The parameter selection (bandwidths and polynomial degrees) for the local regression (where in version 4 the rectangular kernel is replaced by triangular kernel) was made subjectively and independent of the characteristics of the individual time series mainly by looking at the amplitude- and phase functions of the corresponding linear filters. In accordance with these filter functions the parameter selection varies between the interior part and instants in the boundary area of the time series.

A natural next step was the introduction of a broader class of kernels (Heiler and Michels, 1994) according to the idea of locally weighted regression as introduced by Stone (1977) and Cleveland (1979).

With a fixed parameter set local and locally weighted regression with polynomials and trigonometric functions works like a moving average in the interior part of a time series and involves an optimal boundary adaptation property in a mean squared error sense, in contrast to ordinary (Nadaraya–Watson) nonparametric regression where special boundary kernels are necessary in order to avoid a large increase in bias towards the boundary.

Besides its high flexibility, this automatic boundary adaptation property is also a big advantage of a nonparametric time series decomposition method based on locally weighted regression as compared to procedures based on a parametric seasonal ARIMA type model like TRAMO/SEATS (Maravall and Pierce, 1987; Maravall and Gomez, 1994) or X-12 (Findley et al., 1998), where model-based extrapolations on either end of the time series are necessary in order to be able to apply a symmetric filter, derived from the parametric model.

The increasing literature on bandwidth selection in density estimation and nonparametric regression suggests to replace the fixed parameter selections in BV 4 by a data-driven procedure which adapts the decomposition method to the individual peculiarities of a time series.

Besides putting the problem of nonparametric time-series decomposition into the general frame of locally weighted regression, which makes it possible to derive estimates of derivatives and consistency and asymptotic normality results, the main aim of this paper is to adapt procedures developed for bandwidth selection in nonparametric regression to the problem under consideration. For local polynomial regression a procedure based on double smoothing is discussed in Heiler and Feng (1998). A simulation study in Heiler and Feng (1997) shows that a double smoothing procedure with a bootstrap variance estimator performs clearly better than one with a difference-based estimator. A suitable adaptation of this double smoothing bootstrap procedure is suggested for the data-driven decomposition procedure.

Nonparametric approaches to time-series decomposition based on splines with suitably chosen differential operators for the trend-cyclical and seasonality smoothness are discussed in Heiler and Hebbel (1985), Hebbel and Heiler (1987a, b). They can also be used in a local version and yield estimates with comparable smoothness and flexibility properties as the locally weighted regression approach. They will not be discussed in this paper. The problem of selecting the smoothness parameters (and polynomial

degree) is of the same degree of complexity as the parameter selection in the locally weighted regression approach considered in Sections 3 and 4.

The paper is organized as follows. In Section 2 the procedure is presented for a given bandwidth and polynomial degree. Theorem 1 states an asymptotic normality result for the estimation of components. The double smoothing idea is described in Section 3 and in Section 4 the proposed data-driven procedure with the bootstrap variance estimator is presented. In Section 5 the procedure is applied to the quarterly gross domestic product series in Germany as an example. Section 6 contains some concluding remarks concerning comparisons of some existing decomposition procedures.

2. Time-series decomposition with LWR

We consider an equidistant time series Y_t, $t = 1, \ldots, n$. It is assumed that (possibly after some transformation of the original data) Y_t follows an additive components model

$$Y_t = G(t) + S(t) + \varepsilon_t, \quad t = 1, 2, \ldots, n, \tag{2.1}$$

where the ε_t are assumed to be i.i.d. random variables with $E(\varepsilon_t) = 0$ and $\mathrm{var}(\varepsilon_t) = \sigma^2$. G is the trend-cyclical component, S is the seasonal component with seasonal period s and $m := G + S$ is the mean function.

The trend-cyclical component is assumed to be at least $(p+1)$ times differentiable, so that it can be expanded in a Taylor series around a point t_0, yielding a local polynomial representation of order p. In a similar way the seasonal component can be locally modelled by a Fourier series, i.e.

$$G(t) \doteq \sum_{j=0}^{p} \beta_{1j}(t_0)(t - t_0)^j,$$

$$S(t) \doteq \sum_{j=1}^{q} [\beta_{2j}(t_0) \cos \lambda_j (t - t_0) + \beta_{3j}(t_0) \sin \lambda_j (t - t_0)],$$

where $q = [s/2]$ with $[x] = \textit{largest integer} \leqslant x$, $\lambda_1 = 2\pi/s$ is the seasonal frequency and $\lambda_j = j\lambda_1$.

Put

$$\beta_1(t_0) = (\beta_{10}(t_0), \ldots, \beta_{1p}(t_0))^{\mathrm{T}},$$

$$\beta_2(t_0) = (\beta_{21}(t_0), \beta_{31}(t_0), \ldots, \beta_{2q}(t_0), [\beta_{3q}(t_0)])^{\mathrm{T}},$$

$$\beta(t_0) = (\beta_1(t_0)^{\mathrm{T}}, \beta_2(t_0)^{\mathrm{T}})^{\mathrm{T}},$$

$$\boldsymbol{x}_1(t) = (1, (t - t_0), \ldots, (t - T_0)^p)^{\mathrm{T}},$$

$$\boldsymbol{x}_2(t) = (\cos \lambda_1 (t - t_0), \sin \lambda_1 (t - t_0), \ldots, \cos \lambda_q (t - t_0), [\sin \lambda_q (t - t_0)])^{\mathrm{T}},$$

$$\boldsymbol{x}(t) = (\boldsymbol{x}_1(t)^{\mathrm{T}}, \boldsymbol{x}_2(t)^{\mathrm{T}})^{\mathrm{T}}$$

(the last terms of β_2 and $\boldsymbol{x}_2$, respectively, are only necessary for odd s, for even s they have to be omitted) and $\boldsymbol{X} = (\boldsymbol{X}_1 \vdots \boldsymbol{X}_2)$ with the rows $\boldsymbol{x}(t)^{\mathrm{T}}$, where $\boldsymbol{X}$ is the $n \times (p+s)$-regressor matrix.

Let K, the weighting function of LWR, be a second-order kernel with compact support $[-1,1]$, $h \in \mathbb{N}$ be a bandwidth of the observation time, $\boldsymbol{y} = (Y_1, \ldots, Y_n)^{\mathrm{T}}$ denote the data vector and $\boldsymbol{K} = \mathrm{diag}(k_t)$ be a weight matrix with

$$k_t = \begin{cases} K(\frac{t-t_0}{h+0.5}), & t \in [t_0 - h, t_0 + h], \\ 0 & \text{otherwise.} \end{cases}$$

Let $\boldsymbol{e}_j$ denote the jth $(p+1) \times 1$ unit vector, Φ_{s} denote an $(s-1) \times 1$ vector having 1 in its odd entries and 0 elsewhere.

The locally weighted least squares criterion

$$\sum_{t=1}^{n} [Y_t - \boldsymbol{x}_1(t)^{\mathrm{T}} \beta_1(t_0) - \boldsymbol{x}_2(t)^{\mathrm{T}} \beta_2(t_0)]^2 k_t \Rightarrow \min$$

leads to the solutions

$$\hat{\beta}(t_0) = (\boldsymbol{X}(t_0)^{\mathrm{T}} \boldsymbol{K}(t_0) \boldsymbol{X}(t_0))^{-1} \boldsymbol{X}(t_0)^{\mathrm{T}} \boldsymbol{K}(t_0) \boldsymbol{y} \tag{2.2}$$

and

$$\hat{m}(t_0) = (\boldsymbol{e}_1^{\mathrm{T}}, \Phi_{\mathrm{s}}^{\mathrm{T}})(\boldsymbol{X}^{\mathrm{T}} \boldsymbol{K} \boldsymbol{X})^{-1} \boldsymbol{X}^{\mathrm{T}} \boldsymbol{K} \boldsymbol{y} =: \boldsymbol{w}^{\mathrm{T}} \boldsymbol{y}, \tag{2.3}$$

$$\hat{G}(t_0) = (\boldsymbol{e}_1^{\mathrm{T}}, \boldsymbol{0}^{\mathrm{T}})(\boldsymbol{X}^{\mathrm{T}} \boldsymbol{K} \boldsymbol{X})^{-1} \boldsymbol{X}^{\mathrm{T}} \boldsymbol{K} \boldsymbol{y} =: \boldsymbol{w}_1^{\mathrm{T}} \boldsymbol{y} \tag{2.4}$$

and

$$\hat{S}(t_0) = (\boldsymbol{0}^{\mathrm{T}}, \Phi_{\mathrm{s}}^{\mathrm{T}})(\boldsymbol{X}^{\mathrm{T}} \boldsymbol{K} \boldsymbol{X})^{-1} \boldsymbol{X}^{\mathrm{T}} \boldsymbol{K} \boldsymbol{y} =: \boldsymbol{w}_2^{\mathrm{T}} \boldsymbol{y}, \tag{2.5}$$

where $\boldsymbol{0}$ denotes a vector of zeros of appropriate dimension.

$\hat{m}$, $\hat{G}$ and $\hat{S}$ are all linear smoothers and only observations with $t \in [t_0 - h, t_0 + h]$ obtain nonzero weights. For estimations at another instant in the central part of the time series we obtain the same weighting systems (the nonzero parts of $\boldsymbol{w}, \boldsymbol{w}_1$ and $\boldsymbol{w}_2$).

In order to cope with boundary problems we distinguish a left bandwidth h_l and a right bandwidth h_r and put $h = \max(h_l, h_r)$. It is assumed that $h_l < t$, $h_r \leqslant n - t$ and $h_{\mathrm{T}} > p + s$, where $h_{\mathrm{T}} = h_l + h_r + 1$ is called the total bandwidth. If h_l and h_r are fixed and $t \in [h_l + 1, n - h_r]$, then all of $\hat{m}$, $\hat{G}$ and $\hat{S}$ work as moving averages with symmetric weighting systems for $h_l = h_r$ or asymmetric weighting systems for $h_l \neq h_r$.

As in local polynomial fitting the vth derivative of the trend-cyclical component, $0 < v \leqslant p$, can be estimated by

$$\begin{aligned} \hat{G}^{(v)}(t_0) &= v!(\boldsymbol{e}_{v+1}^{\mathrm{T}}, 0^{\mathrm{T}})(\boldsymbol{X}^{\mathrm{T}} \boldsymbol{K} \boldsymbol{X})^{-1} \boldsymbol{X}^{\mathrm{T}} \boldsymbol{K} \boldsymbol{y} \\ &=: (\boldsymbol{w}^{v})^{\mathrm{T}} \boldsymbol{y}. \end{aligned} \tag{2.6}$$

$\hat{G}^{(v)}$ works as a moving average, too.

The estimators $\hat{m}, \hat{G}, \hat{S}$ as well as $\hat{G}^{(v)}$ have properties similar to those of a local polynomial fit as given in Müller (1987). These properties are:

1. Nonzero weights are to be given only to observations within the window $[t-h_1, t+h_r]$ so that the method is local.
2. Polynomials of degree $k \leqslant p$ and an exactly periodic function with the period s are estimated bias free.
3. The weighted sum of squares of the weights is minimized.

Furthermore, the elements of $\boldsymbol{w}$ or $\boldsymbol{w}_1$ sum up to one, and the elements of $\boldsymbol{w}_2$ or $\boldsymbol{w}^1$ sum up to zero.

LWR is an automatic kernel method (Hastie and Loader, 1993). Asymptotically, $\hat{m}, \hat{G}, \hat{S}$ as well as $\hat{G}^{(v)}$ can also be seen as kernel estimators with the special so-called asymptotically equivalent kernels (see Feng, 1998).

In the following we will discuss the asymptotic properties of the proposed estimators in a special case, where the trend function, denoted by $G(x_t)$ with $x_t = (t - 0.5)/n$ is assumed to be a fixed function on $[0, 1]$. Let $S(x_t)$ be a discrete exactly periodic function with period s defined on all observation points $t = 1, 2, \ldots, n$. And let $m(x_t) = G(x_t) + S(x_t)$. Here and in the next section denote by $0 < h < 0.5$ the bandwidth of x. Let $K_p(u)$ be the asymptotically equivalent kernel for $\hat{G}(x_t)$ by a pth order local polynomial fit as defined in Ruppert and Wand (1994). Assume that p is odd and put $k = p + 1$. Furthermore, for a function $f(x)$ define $R(f) = \int f^2(x)\,\mathrm{d}x$. At a point $h < x_{t_0} < 1 - h$ the following theorem holds

Theorem 1. *Under the assumptions*

A1. *$G(x)$ is at least k times continuously differentiable,*
A2. *$h \to 0$ and $nh \to \infty$ as $n \to \infty$,*

we have

(i) *The estimators $\hat{G}(x_{t_0})$ and $\hat{S}(x_{t_0})$ are asymptotically uncorrelated.*
(ii) *The variances of $\hat{G}(x_{t_0})$, $\hat{S}(x_{t_0})$ and $\hat{m}(x_{t_0})$ are given by*

$$\operatorname{var}(\hat{G}(x_{t_0})) = (nh)^{-1}\sigma^2 R(K_p)\{1 + \mathrm{o}(1)\}, \tag{2.7}$$

$$\operatorname{var}(\hat{S}(t_0)) = (nh)^{-1}\sigma^2 (s-1) R(K)\{1 + \mathrm{o}(1)\} \tag{2.8}$$

and

$$\operatorname{var}(\hat{m}(x_{t_0})) = (nh)^{-1}\sigma^2 \{R(K_p) + (s-1)R(K)\}\{1 + \mathrm{o}(1)\}, \tag{2.9}$$

respectively, where K is the weighting function itself.

(iii) *$\hat{S}(x_{t_0})$ is unbiased and the biases of $\hat{G}(x_{t_0})$ and $\hat{m}(x_{t_0})$ are the same*:

$$b(\hat{m}(x_{t_0})) = b(\hat{g}(x_{t_0})) = B_k G^{(k)}(x_{t_0}) h^k [1 + \mathrm{o}(1)] \tag{2.10}$$

with $B_k = (1/(k!))[\int u^k K_p(u)\,\mathrm{d}u]$.

(iv) *If $nh^{2k+1} \to d^2$ as $n \to \infty$,*

$$(nh)^{1/2}(\hat{S}(x_{t_0}) - S(x_{t_0})) \xrightarrow{\mathscr{D}} N(0, \sigma^2 R(K)), \tag{2.11}$$

$$(nh)^{1/2}(\hat{G}(x_{t_0}) - G(x_{t_0})) \xrightarrow{\mathscr{D}} N(\mathrm{d}G^{(v)}(x_{t_0})B_k, \sigma^2 R(K_p)) \tag{2.12}$$

and

$$(nh)^{1/2}(\hat{m}(x_{t_0}) - m(x_{t_0})) \xrightarrow{\mathscr{D}} N(\mathrm{d}G^{(v)}(x_{t_0})B_k, \sigma^2[R(K_p) + R(K)]). \tag{2.13}$$

Proofs of parts (i)–(iii) may be found in Feng (1998). Part (iv) follows from parts (i)–(iii) and Theorem 4.2 in Müller (1988), since all of the weighting systems $\boldsymbol{w}$, $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$ satisfy the conditions of this theorem. The assumption that S is exactly periodic is made here for simplicity but it is not necessary. These results can also be extended to the case where ε_i are correlated errors. Similar results can be obtained for $\hat{G}^{(v)}(x_{t_0})$. Results in Theorem 1 together with $\hat{\sigma}^2$ in Section 4 makes it possible to construct asymptotic confidence intervals for the proposed estimators.

3. Bandwidth selection

In this section we discuss bandwidth selection in local polynomial fitting, which will then be adapted to bandwidth selection for time-series decomposition in the next section. Here $m(x)$ stands for a function on $[0,1]$. The observations are taken at points $x_t = (t - 0.5)/n,\ t = 1,\ldots,n$.

3.1. Procedure based on R statistic

As error criterion for selecting the bandwidth we use the mean-averaged squared error (MASE)

$$M(h) = n^{-1}\sum_{t=1}^{n} E[\hat{m}_h(x_t) - m(x_t)]^2. \tag{3.1}$$

Rice (1983, 1984) proposed an estimator of the MASE, called an R statistic:

$$\hat{R}(h) = n^{-1}\sum_{t=1}^{n} [Y_t - \hat{m}_h(x_t)]^2 - \hat{\sigma}^2 + \frac{2\hat{\sigma}^2}{n}\sum_{t=1}^{n} w_t(x_t), \tag{3.2}$$

where $\hat{\sigma}^2$ is a root n consistent estimator of σ^2. For unbiased $\hat{\sigma}^2$, $\hat{R}$ is an unbiased estimator of the MASE criterion. For σ^2, the difference-based variance estimator $\hat{\sigma}^2$ (Gasser et al., 1986, see also Section 4) may be used. As the bandwidth selected following the cross-validation rule, the bandwidth selector using $\hat{R}$ has a very slow rate of convergence, namely $n^{-1/10}$. It will be only used as a pilot method.

3.2. Double smoothing procedure

The double smoothing (DS) procedure proposed by Müller (1985) has been proved to be a very useful bandwidth selection rule (Härdle et al., 1992; Heiler and Feng,

1997, 1998) which is based on the decomposition of $M(h)$ into a bias part and a variance part,

$$M(h) = V(h) + B(h).$$

It is proposed to estimate the bias at t by means of a pilot smoothing $\hat{m}_g(x_t)$,

$$\hat{b}(x_t) = \sum_{i=1}^{n} w_{ih}(x_t)\hat{m}_g(x_i) - \hat{m}_g(x_t) \tag{3.3}$$

and the bias part of the MASE is then estimated by

$$\hat{B}(h) = n^{-1} \sum_{t=1}^{n} \hat{b}(x_t)^2. \tag{3.4}$$

The variance part of $M(h)$ is estimated by

$$\hat{V}(h) = n^{-1}\hat{\sigma}^2 \sum_{t=1}^{n} \sum_{i=1}^{n} w_{ih}(x_t)^2, \tag{3.5}$$

where $\hat{\sigma}^2$ is also the difference-based variance estimator and $w_{ih}(x_t)$ is the weight of y_i in (2.3) with given bandwidth h. The bandwidth $\hat{h}$ minimizing $\hat{M} = \hat{V} + \hat{B}$ is then defined as the optimal bandwidth. See Härdle et al. (1992) for a slightly different definition of $\hat{M}$.

There are some ways to obtain a data-driven DS procedure. The simplest way is to obtain $\hat{m}_g(x_t)$ in (3.3) with a higher order p_p of the polynomial and the bandwidth selected by the R statistic (Heiler and Feng, 1997). A bandwidth selected by such a simple DS procedure has already much higher rate of convergence than that selected by the R statistic itself. In the case $p = 1$, $p_p = 3$, the rate of convergence of $\hat{h}$ is $n^{-4/9}$, for instance. However, with this simple procedure the highest convergence rate, $n^{-1/2}$, cannot be achieved.

A more complex data-driven DS procedure is to use a so-called factorized pilot bandwidth of the form

$$g = Cn^{\nu}h^{\delta}$$

with properly selected constants C, ν and δ (see Heiler and Feng, 1998). With this method the highest rate of convergence $n^{-1/2}$ can be achieved.

Simulation studies to confirm the finite sample performance of these bandwidth selectors for seasonal time series have been done for nonparametric regression without seasonality (see Heiler and Feng, 1997, 1998 for details) will be reported briefly. The results in both papers show that the double smoothing bandwidth selectors perform much better than those based on the R statistic. It is also shown in Heiler and Feng (1998) that, the root n consistent bandwidth estimator is slightly better than a direct plug-in one. In the sequel, only the first idea will be used because of its simplicity.

4. Data-driven procedure

The R statistic and the DS criterion are both proposed for linear smoothers, which can be easily adapted to obtain data-driven procedures for decomposing seasonal time

series. But in order to develop a data-driven time series decomposition procedure we need a variance estimator for seasonal time series.

4.1. Estimation of σ^2

Rice (1984) proposed a variance estimator:

$$\hat{\sigma}_1^2 = \frac{1}{2(n-1)} \sum_{t=1}^{n-1} (Y_{t+1} - Y_t)^2. \tag{4.1}$$

Furthermore, Gasser et al. (1986) proposed

$$\hat{\sigma}_2^2 = \frac{2}{3(n-2)} \sum_{t=1}^{n-2} \left[Y_{t+1} - \frac{1}{2}(Y_t + Y_{t+2}) \right]^2. \tag{4.2}$$

This idea has been extended by Hall et al. (1990) to a general *difference-based* variance estimator in nonparametric regression with a given *difference sequence* $D_m = \{d_0, d_1, \ldots, d_m\}$, which fulfills

$$\sum_{j=0}^{m} d_j = 0 \quad \text{and} \quad \sum_{j=0}^{m} d_j^2 = 1, \quad m = 1, 2, \ldots . \tag{4.3}$$

The estimation of the variance based on D_m is

$$\hat{\sigma}_m^2 = (n-m)^{-1} \sum_{t=1}^{n-m} \left(\sum_{j=0}^{m} d_j Y_{j+t} \right)^2. \tag{4.4}$$

The estimators in (4.1) and (4.2) are two special cases of $\hat{\sigma}_m^2$ with $m = 1$ and 2, respectively.

In order to estimate the variance in a seasonal time series, Heiler and Feng (1996) proposed a seasonal-difference-based variance estimator by using a *seasonal difference sequence* $D_{ms} = \{d_j \mid j = 0, 1, \ldots, m\}$ in (4.4), which in addition to (4.3) also satisfies the condition

$$S_i = \sum_{j=0}^{m} d_j \delta_{ij} = 0, \quad i = 0, 1, \ldots, s-1,$$

where s is the seasonal period and

$$\delta_{ij} = \begin{cases} 1, & \text{if } (j-i)/s \text{ is an integer,} \\ 0, & \text{otherwise.} \end{cases}$$

For $s = 12$ two simple examples of seasonal difference sequences are

$$D_{14,12} = \frac{1}{c_1}\{-1, 2, -1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, -2, 1\}$$

and

$$D_{26,12} = \frac{1}{c_2}\{-1, 2, -1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, -4, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, -1, 2, -1\},$$

where c_1 and c_2 are two constants defined by (4.3). Similarly one can define D_{ms}^1, $m = s + 2$, and D_{ms}^2, $m = 2s + 2$, for any $s \geqslant 3$. Whereas in $\hat{\sigma}_2^2$ a local linear trend in the

data is annihilated, with $D_{14,12}$ resp. $D_{26,12}$ also a periodic function with period 12 is annihilated. In the sequel, a difference-based variance estimator is denoted by $\hat{\sigma}^2_{\mathrm{G}}$.

By means of the difference-based variance estimator we can obtain a more effective bootstrap variance estimator. According to Heiler and Feng (1997) the bootstrap variance estimator is defined by

$$\hat{\sigma}^2_{\mathrm{B}} = n^{-1} \sum_{i=1}^{n} r_i^2,$$

where the r_i's are the residuals of a data-driven pilot smoothing. This estimator is just the averaged residual sum of squares of a pilot estimate. To calculate $\hat{\sigma}^2_{\mathrm{B}}$ one should use the bandwidth $\hat{h}_v = \mathrm{CF}\hat{h}$ for evaluating the residuals r_i, where $\hat{h}$ is a bandwidth selected for estimating m and CF is a constant, called a bandwidth correction factor. CF can be calculated based on the asymptotic results in Chapter 4 of Feng (1998). For seasonal time series, assume that the bisquare kernel is used as weight function. Then we have $\mathrm{CF} = 1.3188$ for $s = 4$ and $\mathrm{CF} = 1.3257$ for $s = 12$, if the residuals r_i, $i = 1, \ldots, n$, are obtained from a local smoothing with $p = 3$. To calculate $\hat{\sigma}^2_{\mathrm{B}}$ an iterative procedure is proposed according to Heiler and Feng (1997) using the k-NN (k-nearest neighbour) estimators. That is $h_{\mathrm{T}} = k$ is fixed at each point t. h_l and h_r are taken as close to each other as possible. Let $\hat{k}_i$ be the selected bandwidth for estimating m with $p = 3$ and $\hat{k}_{v,i} = [\mathrm{CF}\hat{k}_i]$ be the corresponding bandwidth used for calculating $\hat{\sigma}^2_{\mathrm{B},i}$ in the ith iteration, respectively. Starting with "$i = 0$", the algorithm proceeds as follows:

1. If $i = 0$, calculate $\hat{\sigma}^2_{\mathrm{G}}$ and put $\hat{\sigma}^2_{\mathrm{B},0} = \hat{\sigma}^2_{\mathrm{G}}$. Otherwise, let $\hat{k}_{v,i} = [\mathrm{CF}\hat{k}_i]$ and calculate $\hat{\sigma}^2_{\mathrm{B},i}$;
2. Select the bandwidth for $p_p = 5$ by the R statistic with variance $\hat{\sigma}^2_{\mathrm{B},i}$ and fit $\hat{m}_g$;
3. Select the bandwidth $\hat{k}_i$ following the DS criterion with $p = 3$, $\hat{m}_g$ and $\hat{\sigma}^2_{\mathrm{B},i}$. If $i > 0$ and $\hat{k}_i = \hat{k}_{i-1}$, put $\hat{\sigma}^2_{\mathrm{B}} = \hat{\sigma}^2_{\mathrm{B},i}$. Otherwise, put $i := i + 1$ and go back to step 1.

The simulation in Heiler and Feng (1997), where nonparametric regression without seasonality was considered, showed that $\hat{\sigma}^2_{\mathrm{B}}$ performs much better than $\hat{\sigma}^2_{\mathrm{G}}$. The above-mentioned simulation also shows that the former of $\hat{\sigma}^2_{\mathrm{B}}$ adapts automatically to the variation in a given sample. Hence, the bandwidth based on $\hat{\sigma}^2_{\mathrm{B}}$ is not too far from the true bandwidth even $\hat{\sigma}^2_{\mathrm{B}}$ is far from σ^2. See Heiler and Feng (1997) for details.

4.2. The procedure

The data-driven procedure consists in two steps. In the first step of this procedure, for the central part of the time series with a symmetric neighbourhood an optimal pair $\hat{p}$ and $\hat{h}_m(\hat{p})$ minimizing $\hat{R}$ is selected. It is well known (see e.g. Fan and Gijbels, 1996) that for a symmetric neighbourhood and a symmetric kernel a polynomial of given even degree p and the one with degree $p + 1$ (odd) yield the same weighting system. Hence for a given $p_{\max}$ the search starts with a p_m ($p_m = p_{\max}$, if $p_{\max}$ is even, or $p_m = p_{\max} - 1$, if $p_{\max}$ is odd), on a grid going from $h_{\min} = [\frac{1}{2}(s + p)] + 1$ to

Table 1
Selected $\hat{h}_1$, $\hat{h}_T$ and $\hat{p}$

Observ. time	Parameters selected by R			Parameters selected by DS		
(t)	$\hat{h}_1$	$\hat{h}_T$	$\hat{p}$	$\hat{h}_1$	$\hat{h}_T$	$\hat{p}$
n	8	9	1	8	9	1
$n-1$	13	15	2	15	17	2
$n-2$	12	15	3	15	18	3
$n-3$	12	16	3	15	19	3
$n-4$	12	17	3	15	20	3
$n-5$	12	18	3	15	21	3
$n-6$	10	17	3	16	23	3
$n-7$	9	17	2	17	25	3
$n-8$	11	20	2	18	27	3
$n-9$	10	20	2	11	21	2
$n-10$	10	21	2	11	22	2
$n-11$				11	23	2

$h_{\max} = [n/4]$. After evaluating $\hat{R}(h_m(p))$ and selecting $h_m(p)$ the procedure is repeated with $p-2$ as long as $p-2 \geqslant 0$. From among the $h_m(p)$ with $p = p_m, p_m - 2, \ldots,$ the pair $\hat{p}$ and $\hat{h}_m(\hat{p})$ with smallest $\hat{R}(\hat{h}_m(\hat{p}))$ is selected for the central part. It is clear that an estimate of the MASE should not be smaller than $\hat{V}$. Hence, if $\hat{R} < \hat{V}$ we put $\hat{R} = \hat{V}$.

In the second step of this procedure an optimal pair $\hat{p}$ and $\hat{h}_l$ (number of left neighbours) for each point in the right boundary area is selected, where after the first step the right boundary area is given by the integers in $[n - \hat{h}_m + 1, n]$. For $t \in [n - \hat{h}_m + 1, n]$, the number h_r of right neighbours is restricted by $n - t$. We always put $h_r = n - t$, so that the total number $\hat{h}_T$ of observations included in the local regression at t is given by

$$h_T = n - t + \hat{h}_l + 1. \tag{4.5}$$

In the boundary area all polynomial degrees from $p_{\max}$ down to zero are taken into consideration. These bandwidths are then symmetrically used on the left boundary.

5. An example

The decomposition results for the time series of the quarterly gross domestic product (GDP) in Germany from 1968 to 1994 (billion DM, at current prices) are given here as an example. In this example the bisquare kernel is used as weight function. The left bandwidths $\hat{h}_l(t)$ and the orders of the polynomials $\hat{p}(t)$ (with $p_{\max} = 3$) for $\hat{m}(t) = \hat{G}(t) + \hat{S}(t)$ selected by the R statistic and by the DS procedure, respectively, are given in Table 1 (listed from the right end point $t = n$ to the central part). In the pilot smoothing for the DS procedure $p_p = 5$ was used. The total bandwidths, $\hat{h}_T$, are also given in Table 1. Fig. 1 shows the data-driven decomposition results by the DS procedure. See Heiler and Feng (1996) for more examples, where always the variance estimator σ_G^2 was used.

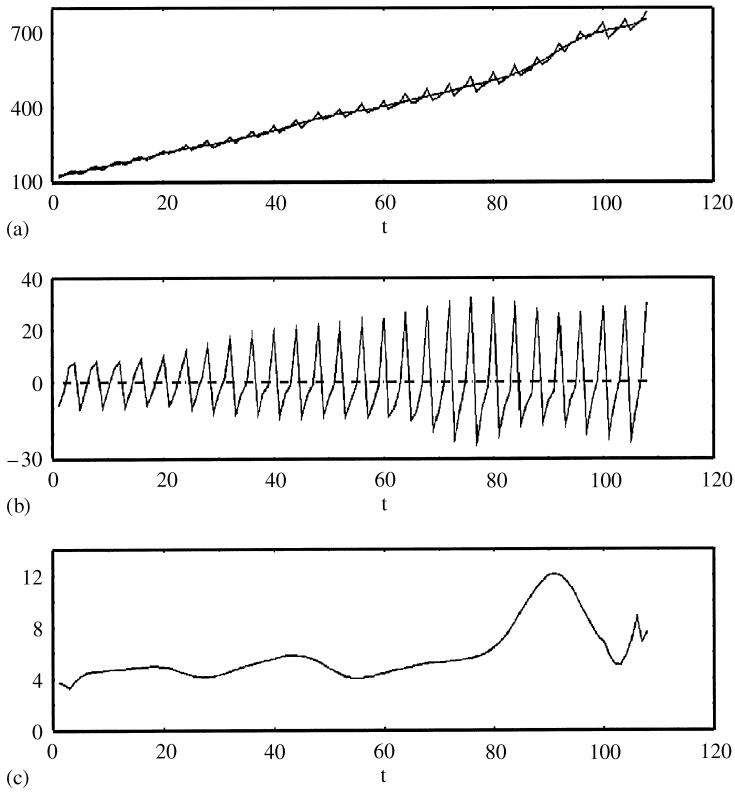


Fig. 1. Decomposition results of the time series GDP using the DS criterion: (a) the data and $\hat{G}$, (b) $\hat{S}$ and (c) $\hat{G}'$.

From Table 1 we see that at the end point a lower polynomial order should be used due to the very large variance in $\hat{m}(n)$ at this point. There is another interesting difference between estimations in the central part and at an end point. Whereas Müller (1987) showed that the optimal weight function in the central part is the Epanechnikov kernel (Epanechnikov, 1969), Cheng et al. (1997) showed that the triangular kernel is the optimal weight function at an end point.

6. Concluding remarks: what is the best method?

The results of decomposition procedures are used in practice mainly in a data-analytic way. Aspects of statistical inference play only a subordinate role. (For direct statistical forecasting different methods are called for anyway). Assessments and comparisons of different methods are difficult since the assumed components are not observable.

Simulation studies are not very helpful in this respect, since it is well known that artificially generated model courses usually favour one particular method. Hence comparative empirical studies conducted in practice are mainly based on heuristic arguments.

Recently Eurostat (1998) carried out a comparison of the main different available computer packages X-12 (Findley et al., 1998), TRAMO/SEATS (Maravall and Pierce, 1987; Maravall and Gomez, 1994), STAMP (Harvey and Todd, 1983) and BV 4. The first two methods are based on parametric seasonal ARIMA models, the third starts from a structural state space mode. The study is mainly confined to the comparison of the different underlying models and the details of the packages (where for the people from Eurostat a nonparametric model is not a model). Comparisons with empirical macroeconomic time series were carried out by the German Federal Statistical Office (Höpfner, 1998). As one obvious feature the nonparametric procedure yields smoother courses for the estimate of the trend-cyclical movements and a higher flexibility of seasonal movements. This fact holds also true if the data-driven version discussed in this paper is used. For practitioners the behaviour at the actual boundary and the amount of revisions of evaluated courses when new data come in play an important role. As a matter of fact the two ARIMA model-based procedures X-12 and TRAMO/SEATS react very sensitively to outliers among the very recent data, since they use forecasts based on a seasonal ARIMA model in order to be able to apply a symmetric filter in the boundary area. Such forecasts can be badly spoiled in the presence of outliers. In such cases subjective corrections are necessary. The re-evaluations of the parametric models when new data come in lead to alterations of the course of estimated components over the whole observation period, whereas for the nonparametric approach revisions occur only in the boundary area (with length according to the estimated bandwidth).

Although there is no uniformly best procedure for decomposing seasonal time series which meets all needs of practitioners in an ideal way we think that a nonparametric data-driven procedure based on the ideas discussed in this paper is a good candidate from a theoretical as well as a practical point of view.

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